

Diabetic Retinopathy Early Diagnosis(DL + Blockchain Pipeline)

Introduction:

Diabetic retinopathy is an eye condition that can cause vision loss and blindness in people with diabetes. Early diagnosis can help ensure timely treatment, because the disease often develops silently with no noticeable symptoms. For early diagnosis, we use a Deep Learning model (EfficientNetB3) to classify retinal images by DR severity, and store the results on the blockchain for tamper-proof records with role-based access control.

Dataset:

For fine-tuning EfficientNetB3, we used the APTOS 2019 dataset having 2,930 training images across 5 severity categories (No DR, Mild, Moderate, Severe, Proliferative) mapped as classes 0–4. The dataset was heavily imbalanced (No DR had 1,434 images while Severe and Proliferative had only 154 and 234 respectively). We augmented Mild, Severe, and Proliferative to reduce this imbalance. Custom class weights were also assigned, Mild (3.0) and Moderate (3.0) to prioritize early diagnosis, and Severe (2.0) and Proliferative (2.0) for dangerous late-stage cases.

Deep Learning Model:

For classification, we used EfficientNetB3 pretrained on ImageNet. Transfer learning was chosen because training from scratch would require a much larger dataset. Training was done in two phases. In Phase 1, the base model was frozen and only the custom classification head was trained consisting of GlobalAveragePooling2D, Dropout(0.3), and a Dense layer with 5 outputs and softmax activation. In Phase 2, the last 30 layers of the base model were unfrozen for fine-tuning at a lower learning rate (1e-5), allowing the model to adapt deeper features to retinal images. Both phases used Adam optimizer, sparse categorical crossentropy loss, EarlyStopping (patience=5), and ReduceLROnPlateau to prevent overfitting.

The final model achieved 79% overall accuracy on the validation set (586 images). Mild and Moderate classes showed reasonable recall (0.60 and 0.82), which aligns with the early diagnosis goal. Recall for Severe (0.17) and Proliferative (0.29) remains low, a known limitation discussed below.

Smart Contract:

The smart contract stores patient records, assigned doctors, diagnosis predictions, and the SHA-256 hash of the retinal image used for each diagnosis. Role-based access control is enforced on-chain, only the hospital admin can register patients and assign doctors, only the assigned doctor can upload a patient's diagnosis, and doctors can view records of their own patients while the admin can view all records. The contract is deployed on the Sepolia testnet.

Python Bridge (Web3.py):

A synthetic patient profile is generated using Faker, registered on-chain via the smart contract, and saved to a local JSON store. A test image is randomly selected, passed through the model for DR classification, assigned to a patient, and marked as used to prevent data leakage. The image hash and predicted class are then uploaded to the blockchain. Tamper detection is also implemented, on-chain data can be verified at any time by fetching the stored hash, recomputing the image hash locally, and comparing both.

End-to-End Pipeline:

Register Patient → Pick Image → Model Predict → Hash Image → Upload to Blockchain → Verify

Limitations:

- **Doctor Authentication:** Admin and doctor share the same wallet in this PoC. In production, each doctor would hold a separate wallet with their own private key.
- **Patient Data Storage:** Patient data is stored in a local JSON file. A production system would require an encrypted database with blockchain-enforced access control at the API layer.
- **Model Recall:** Recall for Severe (0.17) and Proliferative (0.29) classes is low. These are the most critical cases to detect, further fine-tuning, focal loss, or additional data collection is needed.
- **No Frontend:** The pipeline is entirely script-based. A clinical deployment would require a proper web interface.